

SLM to control laser irradiation

Hamamatsu Photonics has developed a liquid-crystal spatial light modulator (SLM) having a high-pulse laser power capability of up to 400GW/cm².

Through this development, innovative manufacturing and fabrication techniques can be carried out by using a cyber-physical system (CPS) that combines cyberspace (virtual space) and physical space (real space) to a highly advanced degree. SLM can freely control the irradiation pattern of a laser.



By incorporating an SLM into an industrial pulsed laser system, the irradiation of diffracted multiple laser beams is allowed onto different points of the target sample simultaneously. This improves the processing throughput compared to laser machining that uses only a single focused beam.

Hamamatsu Photonics
www.hamamatsu.com

Speech synthesis ICs optimised for sound

With the increasing presence of ADAS and other safety systems designed to notify drivers of vehicle operation and the surrounding environment, auto manufacturers have now been also prompted to develop audio systems that utilise sound effects and voice to issue warnings and alerts in order to maintain the safety of vehicle occupants and pedestrians.

Keeping this in mind, LAPIS Semiconductor has introduced the ML2253x series of automotive speech synthesis

ICs that have been optimised for audible notifications and sound effects in the Advanced Driver Assistance System (ADAS) and Acoustic Vehicle Alerting System (AVAS).



The new product series not only improves quality by detecting audio defects using a playback anomaly function but also facilitates system configuration while simplifying software design for the main MCU, claims the company. The ML2253x series also includes a playback sound anomaly function that can send error signals to the main MCU.

LAPIS Semiconductor
www.lapis-semi.com

Carbon-based ultracapacitors

The next-generation ultracapacitors are set to revolutionise the growing global market for the Internet of Things (IoT) devices by offering huge benefits such as cost, efficiency, safety, and environmental impact.



However, there are major challenges around the performance of IoT devices, raising the question of how to power these devices efficiently, cost-effectively, and with minimal environmental impact.

Given this, NAWACap ultracapacitors have been launched by NAWA Technologies that are based on carbon. The NAWACaps can be packaged into small, flat cells and withstand extreme temperatures and harsh environments, making them an ideal match for IoT

sensors that have to cope with high current peaks. Another benefit is that these capacitors come with a long life and therefore do not need to be replaced.

NAWA Technologies
www.nawatechnologies.com

Baseplate-cooled power supply

Here's a highly compact 230W innovative baseplate-cooled power supply for medical, household, industrial, and ITE applications. The new, compact RACM230-G series power supplies from



RECOM support up to 160W continuous output power and 230W for several seconds for loads with peak demands, without fan cooling. With forced air, depending on the model, line input, and ambient temperature, 230W is available continuously.

Outputs available are 12V, 24V, 36V, 48V, and 54V nominal with conversion efficiency up to 92 per cent and an operating temperature range of -40 degrees Celsius to +80 degrees Celsius. Standby power consumption is less than 500mW. For flexibility, the RACM230 series is available as an open-frame design or in an enclosed metal housing.

RECOM
www.recom-power.com

World's first flip-type capacitors

TDK Corporation has introduced a new capacitor series, which it claims to be the world's first flip-type MLCCs (multilayer ceramic chip capacitors) in 0510 design (EIA 0204). These are suited for applications involving Advanced Driver Assistance System (ADAS) and other automated driving technologies. As per the press release, the miniature components have voltages between 4V and 50V